AUTOMATED MORPHOMETRIC READ-OUTS OF THE NEURITE NETWORK IN A CARDIAC INNERVATION CHIP WITH A COMPACT ATTENTION U-NET

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ABSTRACT

Sympathetic neurons in a cardiac innervation chip build a flat neurite network, whose fragmentation under cardiotoxic stress can serve as a safety read-out. To trace this network by hand takes hours per image; classical vessel filters fail where fibres cross each other; furthermore, obtaining high-quality images is difficult. We describe a pipeline that takes a raw confocal stack as input and outputs seven morphometric read-outs; we evaluate it out-of-fold on 15 annotated images. Seven different U-Net models have at most 1.2 Dice points difference, and the deployed Attention U-Net (8M parameters) has an out-of-fold Dice of 0.70, which is higher than the Dice of Frangi (0.56) and Otsu (0.42) on all images, and close to that of nnU-Net (0.72, p=0.12). On two images from a new batch, total fibre length differs by less than 6% from manual tracing. Length-based read-outs can be reported directly, while junctions and branches can be quantified after setting an uncertainty range. This pipeline lets biologists measure neurite morphology without manual tracing.

***Index Terms*—** neurite segmentation, U-Net, morphometrics, cardiac innervation chip, cardiotoxicity

1. INTRODUCTION

Cardiotoxicity is a major cause of drug failure in development or removal from the market [1, 2]; therefore human stem cell models of the heart are increasingly used to detect cardiotoxicity early in development [3, 4]. Because cardiac function is regulated by sympathetic innervation, toxic effects may also manifest in associated neuronal networks. The cardiac innervation chip used in this study is populated with sympathetic neurons derived from human induced pluripotent stem cells (hiPSCs), which grow their neurites through micro-channels into a separate compartment, where the neurites form a flat network (Fig. 1a). In response to cardiotoxic stress the network breaks up and retracts, and thus the morphology of the network could be a promising image-based measure of damage. We aim to extract the length, branching and coverage of this network directly from a raw confocal stack without manual tracing.

This is challenging because the fibres are thin, faint, and highly overlapping, while the read-outs are sensitive to network topology: even a single missing pixel at a crossing can change the number of detected junctions. The Simple Neurite Tracer (SNT) [5] can deal with this semi-automatically, but it takes hours to trace one 2048×2048 image, and vesselness filters [6, 7] are fast, but they also fail at those crossings and branch points. There is also a lack of labelled data. Each image takes several weeks of culture on a chip and hours of tracing, which means that a study may only end up with about 15 high-resolution images, and with so little data it is easy to overestimate the accuracy when the same images are used for training and testing. Finally, the end users are biologists, who need descriptive metrics rather than masks, and a high segmentation score alone does not indicate how reliable those metrics are.

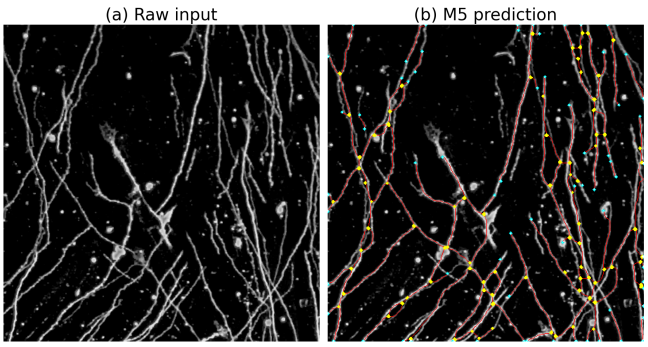


Fig. 1. The task. (a) Preprocessed confocal image of the neurite compartment. (b) Automatic result: network centre-line (red), junctions (yellow) and endpoints (cyan). The seven morphometric read-outs are computed from this graph.

In a controlled ablation we found that with this much data, changing the network does not really make any difference: adding attention gates, making the encoder twice as wide (four times the parameters), or changing to a topology-aware loss function did not change the accuracy. We therefore opt for a small Attention U-Net [8, 9] and put most of the effort into the rest of the pipeline (Fig. 2), which includes preprocessing optimized for chip images, an ensemble of the five fold models, and a skeleton graph analysis which yields seven read-outs. The full pipeline is available as a desktop application which requires no programming. All of the accuracy figures we highlight are out-of-fold or computed on a separate batch, and we specify how reliable the read-outs are.

Out-of-fold, the Dice of the deployed model on the full image is 0.70; it is superior to both Frangi and Otsu on all 15 images, and the difference compared to a default nnU-Net v2 [10] is not significant. On two new images from a different batch, the centre-line Dice is 0.84, and the total fibre length is within 6% of manual tracing. The main contributions of this paper are:

- to our knowledge, the first automated pipeline for this cardiac innervation chip that goes from raw `.oir` stacks to seven morphometric read-outs, packaged for use by biologists;
- an ablation of seven U-Net variants which shows that, with 15 high-resolution images, architectural changes have no measurable effect;
- an out-of-fold comparison with classical filters and nnU-Net, together with a test on images from a new batch;
- an analysis of which read-outs need an uncertainty range.

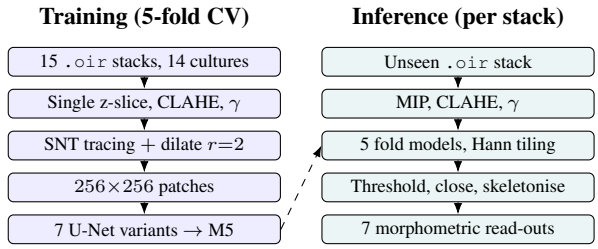


Fig. 2. Training and inference pipelines. The five fold models of M5 are used together as the inference ensemble (dashed arrow). Training uses one in-focus z-slice per stack; inference uses a maximum-intensity projection (MIP).

2. MATERIALS AND METHODS

2.1. Data and annotation

The chips were imaged with an Olympus FV4000 confocal microscope ($10\times$, $0.6215\,\mu\mathrm{m\,px}^{-1}$, 2048×2048 pixels, 14–20 z-slices). After a quality check, 15 of the 33 acquisitions were kept, each represented by one in-focus z-slice. These come from 14 cultures, as two slices belong to the same culture; merging these two in the paired tests does not change any conclusion. We correct each image for uneven illumination (rolling ball, gamma), remove noise (despeckle, median filter) and apply CLAHE [11], which brightens faint distal fibres while dense bundles do not saturate. The ground truth was traced in SNT for every image. A trace of one pixel wide would punish a one-pixel shift as heavily as a missing fibre, so we dilate the traces with a disk of radius 2 px and use these as training masks.

2.2. Network and ablation

We use an Attention U-Net [9] for segmentation; U-Net was designed for training on few annotated images [8], and its skip connections preserve thin structures. Our deployed model, M5, has encoder widths $[32, 64, 128, 256]$, with 8M parameters. To understand what design choices are important, we trained seven variants, each varying in only one aspect (encoder width (8M vs 31M parameters), attention gates, topology-aware clDice loss [12], elastic augmentation; Table 1). Each variant uses the same five image-level folds with the same seed, so each is trained on 12 images. We cut images into 256×256 patches using foreground-aware sampling and apply augmentations on the fly. We use Adam with a 10^{-4} learning rate and cosine decay, a weighted binary cross-entropy and Dice loss, and early stopping.

2.3. From mask to morphometric read-outs

At inference time, the z-stack is reduced to a maximum-intensity projection (MIP), so there is no need to pick a slice by hand. Each of the five fold models is run with overlapping tiles and blended using a Hann window [13] to avoid visible seams, and the five probability maps are averaged. The average map is thresholded at 0.70, closed with a disk of radius 2 px and skeletonised [14], and branches of length less than 8 px are removed, to avoid artefacts giving false junctions. The skeleton is then considered as a graph, with endpoints having one neighbour and junctions having three or more. Seven read-outs are calculated from this graph, namely total fibre length, number of junctions, number of branches, mean branch length, branchiness (junctions per micrometre), covered area and mean thickness.

Table 1. Ablation over seven U-Net variants (5-fold cross-validation, patch-level DSC; higher is better).

Model	Width	Attn.	clDice loss	Params	DSC ↑
M1 Baseline	wide	–	–	31M	0.706
M2 +Elastic aug.	wide	–	–	31M	0.716
M3 +Attention	wide	yes	–	31M	**0.718**
M4 +clDice	wide	–	yes	31M	0.716
M4b Attn.+clDice	wide	yes	yes	31M	0.709
M5 (deployed)	light	yes	–	8M	0.716
M6 Light, no attn.	light	–	–	8M	0.717

2.4. Evaluation protocol

Since the dataset is small, we present three types of evaluation: out-of-fold (OOF), held-out, and in-sample. OOF refers to the strictest evaluation: each image is segmented by the fold model that did not see this image during training. OOF is our main measure of accuracy, and nnU-Net is evaluated the same way on the same splits. The held-out results are based on two acquisitions from a later batch, which were traced from scratch without looking at the prediction; they were evaluated on MIP input, restricted to the imaged wells where the manual tracing was made. The in-sample results refer to the prediction of the five-fold ensemble on the 15 development images; these show agreement with the annotator, but tell nothing about the generalisation behaviour. We report the Dice score (DSC) and the centre-line Dice (clDice) [12], computed with a 2-pixel matching tolerance (equal to the dilation radius). Since M5 was trained without the clDice loss, this metric is independent of its training objective.

3. RESULTS

3.1. Effect of the architecture

The patch-level DSC for all seven variants is between 0.706 and 0.718 (Table 1), although the number of parameters differs by a factor of four; adding attention gates changes DSC by at most 0.002 (M2 vs M3 and M6 vs M5), and the small encoder is as effective as the wide one. There is no gain from the clDice loss, likely because the dilated masks already encourage connected predictions. We deployed M5, the lightweight version of the best wide model (M3); since M5 and M6 (no attention) perform identically, this choice does not affect the results.

3.2. Comparison with baselines

Table 2 and Fig. 3 show the out-of-fold results; the full-image DSC of M5 is 0.702 ± 0.089, in line with its patch-level score. M5 scores higher than Frangi and Otsu on each of the 15 images (paired Wilcoxon $p < 0.001$), with a median margin of $+0.15$ over Frangi. Compared with a default 2D nnU-Net v2 trained on the same folds, the median difference is -0.013 (95% bootstrap CI $[-0.033, 0.007]$, p=0.12), and M5 is better on 5 of the 15 images (Fig. 3). So at this data size the small network performs about as well as the self-configuring baseline, but not better. The most difficult image, which contains only a few faint fibres with low contrast, is also the most difficult one for all other methods. In Fig. 4, Otsu marks too much

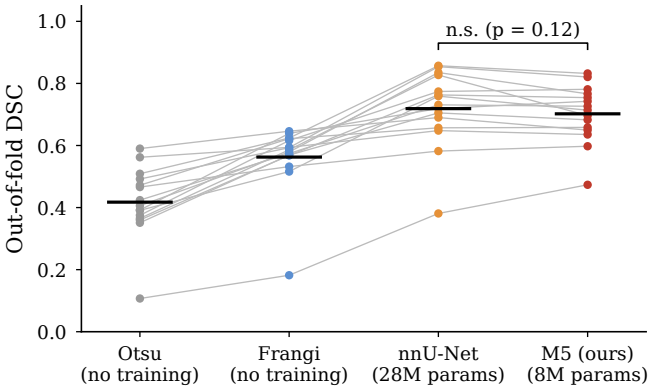


Fig. 3. Out-of-fold DSC per image for each method; grey lines connect the same image and black bars are the means. M5 beats Otsu and Frangi on every image. The difference with nnU-Net is not significant (p=0.12), although M5 has about 3.5 times fewer parameters (8M vs. 28M).

Table 2. Segmentation accuracy (mean±SD over images; higher is better). OOF: each image is segmented by a model that never saw it. clDice uses a 2-pixel tolerance.

Protocol	Method	DSC ↑	clDice ↑
OOF (n=15)	Otsu	0.418 ± 0.109	–
	Frangi	0.563 ± 0.108	–
	nnU-Net v2	$\mathbf{0.719} \pm 0.123$	–
	M5 (ours)	0.702 ± 0.089	0.795 ± 0.089
Held-out (n=2)	M5 ensemble	–	0.84
In-sample (n=15)	M5 ensemble	0.887 ± 0.020	0.96 ± 0.02

background and Frangi breaks up at crossings, whereas the predicted mask follows the manual tracing. The in-sample ensemble reaches 0.887 ± 0.020, which we only use as a measure of agreement with the annotator.

3.3. Images from a new batch

We also evaluated M5 on two acquisitions from a subsequent batch that were not used for training, model selection or tuning. In this test, M5 achieves a clDice of 0.84 within the imaged wells, and the total fibre length on both images is within 6% of the manual measurement ($+2.4\%$ and -5.6%). The images are different from the training images in two respects: they are from a different batch, and they were scored on MIPs, whereas M5 was trained on single slices. The test covers both types of change; however, given that only two images are available, we can only give a rough indication, and not an accurate estimate, of the generalisation error.

3.4. Reliability of the read-outs

Our results suggest that length-based read-outs can be reported directly, while counts need an uncertainty range; we base this on two separate analyses. In a Bland–Altman analysis [15] against the manual tracings (15 development images, in-sample ensemble), total fibre length has a bias of $+0.3\%$ and 95% limits of agreement of roughly -10% and $+11\%$. For the number of junctions the bias is $+5\%$ and the limits are roughly -30% and $+40\%$. Between the five fold models, the length-based read-outs vary little: the coefficient of variation is 8.5% for total length, 10.6% for branchiness and 12.3% for mean branch length, and the counts vary about twice as much (19.1% for junctions, 20.8% for branches), since a single gap at a crossing adds or removes a junction. For this reason we report the counts together with their spread over the fold models. Lastly, we ran the pipeline on a new batch of eight chips with three Nerve Growth Factor conditions; it produced a complete profile for every chip without manual intervention. This batch is too small to draw biological conclusions from, so we only see it as a test of the deployed tool.

4. DISCUSSION AND LIMITATIONS

For a lab with 15 annotated high-resolution images, we find that it is worth investing time in annotation, preprocessing and a measurement step that reports its own uncertainty, rather than in a large network. Attention, width and a topology-aware loss had no measurable effect, and the small network performed as well as nnU-Net. There are five limitations to these results. *(1)* Size of the data set. We only had 15 images from 14 cultures, so the ablation cannot detect small effects and the overall out-of-fold (OOF) estimate has a large standard deviation (SD 0.09); we will be annotating more chips and splitting the data by chip rather than by image. *(2)* Single annotator. All our metrics are calculated by comparing to a single annotator. A second annotator on a stratified subset would show the ceiling in inter-annotator agreement. *(3)* Tuning on the development set. We chose the threshold, closing radius and pruning length on the 15 development images. Our sensitivity curves were flat (within 0.01 DSC), and the held-out images were evaluated with these same values. *(4)* Slice to MIP. We only evaluated the change from training on single slices to testing on MIPs on 2 images. We trained on single in-focus slices because they give the sharpest fibre boundaries for manual tracing, while at deployment the MIP captures the whole network without manual slice selection. *(5)* Comparison with nnU-Net. We used the default 2D plan with a shorter schedule and without our preprocessing. We did not compare with a deep learning model that has been made specifically for neurites.

5. CONCLUSION

We developed an automated pipeline that takes raw confocal stacks of a cardiac innervation chip and returns seven morphometric read-outs, with no programming required. As the architecture matters little at this data size, we use a small Attention U-Net, combined with careful preprocessing, strict evaluation, and a reliability rating of each read-out. In out-of-fold evaluation, our method outperforms classical filters, reaches performance comparable to nnU-Net, and can process images from a new batch. As our next step, we plan to collect a larger dataset containing drug-exposed chips, split data at the chip level, and add a second annotator. The code and desktop application are available on request at `https://github.com/PepBaggen/neurite-segmentation`.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study used human induced pluripotent stem cell lines derived with informed consent under approved protocols; no human participants or animals were involved in the experiments reported here, and no further ethical approval was required.

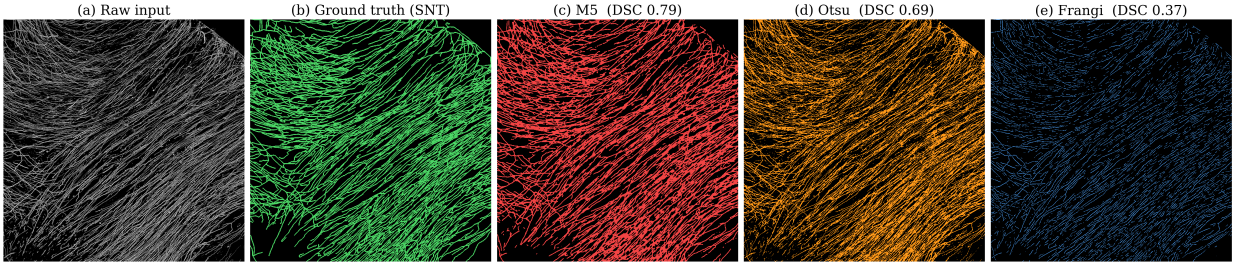


Fig. 4. Segmentation masks on a held-out control well, shown without the grayscale background. (a) Raw input. (b) Manual SNT tracing. (c) M5. (d) Otsu. (e) Frangi. The DSC values in the panels refer to the crop shown.

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